
Start-State Critic Bias: A Per-Update Diagnostic for Actor–Critic Deep RL

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Abstract

1 Actor–critic deep reinforcement learning (RL) relies on learned value functions to
2 guide policy updates. Practitioners typically monitor the Bellman residual (ϵ_u), or
3 value-function loss, as a proxy for critic quality and update health. We demonstrate,
4 both theoretically and empirically, that across the actor–critic configurations we
5 audit the absolute Bellman residual is uninformative as a per-update predictor of
6 policy improvement. Theoretically, we show that the relevant criterion is a *dimensionless ratio* ϵ_u/A_k , and that any fixed absolute tolerance eventually becomes
7 vacuous as the policy improves. Empirically, we audit over 86,000 actor–critic
8 updates across PPO, SAC, and TRPO, spanning eight environments and 237 seeds.
9 We find ϵ_u is near-chance at predicting policy harm (pooled median AUROC 0.46).
10 The exact policy-improvement identity $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$ exposes the
11 **start-state critic bias** $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ as the operative covariate-shift quantity.
12 The bias term predicts harm at pooled median AUROC 0.68 and dominates
13 ϵ_u on 234 of 236 paired seeds (paired Wilcoxon $p < 10^{-40}$). To rule out return-
14 estimate leakage, we replicate the diagnostic on independent held-out rollouts and
15 recover AUROC 0.61, confirming the signal reflects critic generalization rather
16 than shared Monte Carlo noise. The dominance is uniform across PPO, SAC, and
17 TRPO, robust to a ~ 200 -cell hyperparameter factorial, and matches a tabular sanity
18 check where the identity holds to machine precision. Finally, directly targeting the
19 bias via a ν -weighted critic loss leaves the lag-1 AUROC at chance, suggesting
20 the boundary between retrospective diagnostic and prospective controller is fundamental to the optimization order. We conclude that critic monitoring in actor–critic
21 deep RL should shift from absolute TD error toward relative, distribution-aware
22 quantities.
23
24

25 1 Introduction

26 Actor–critic algorithms in deep reinforcement learning (RL)—including PPO [Schulman et al., 2017],
27 SAC [Haarnoja et al., 2018], and TRPO [Schulman et al., 2015a]—update a policy iteratively against
28 a learned value function. The standard analytic guarantee for the success of these updates typically
29 relies on a bound of the form:

$$J(\pi_{k+1}) - J(\pi_k) \geq A_k - c \cdot \epsilon_u, \quad (1)$$

30 where ϵ_u is a norm of the Bellman residual $T^\pi V - V$ and A_k is the advantage signal. In practice,
31 the scalar ϵ_u is the value-function loss minimized by nearly every deep actor–critic implementation
32 [Huang et al., 2022, Andrychowicz et al., 2021], and is the primary quantity practitioners monitor to
33 assess whether their critic is “good enough” to support policy improvement.

34 In this paper, we show that this reliance on the absolute Bellman residual is misplaced. We argue
 35 that ϵ_u is the wrong object for monitoring update health, both because it lacks the correct scale and
 36 because it targets the wrong distribution.

37 **Theory: fixed absolute tolerances eventually become vacuous.** We first give a constructive
 38 argument for why a fixed absolute Bellman-residual tolerance cannot certify policy improvement.
 39 We show that the relevant theoretical tolerance is the *dimensionless ratio* ϵ_u/A_k . As training
 40 progresses and the policy approaches a local optimum, the advantage signal A_k naturally tends
 41 toward zero. Consequently, any fixed absolute tolerance ϵ^{fixed} eventually becomes too large relative to
 42 the available improvement signal, rendering the improvement guarantee (1) vacuous. We formalize
 43 this in Proposition 2, showing that fixed-tolerance critics eventually fail to certify even genuine
 44 improvements.

45 **Mechanism: the start-state bias as covariate shift.** The failure of the Bellman residual to track
 46 improvement is explained by an exact algebraic identity (Proposition 1):

$$J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k, \quad \hat{\Delta}_k = J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]. \quad (2)$$

47 We call $\hat{\Delta}_k$ the **start-state critic bias**: it is the post-update critic’s error *at the initial-state distribution*
 48 ν . The Bellman residual ϵ_u measures critic error on the rollout distribution d^{π_k} ; it is silent about
 49 accuracy at ν . The two distributions disagree even on-policy because d^{π_k} is the discounted state-
 50 visitation distribution and ν is its starting point. The diagnostic failure is then mechanically clear: ϵ_u
 51 measures the right kind of error on the wrong distribution.

52 **Empirical audit across PPO, SAC, and TRPO.** We validate these theoretical insights with an
 53 audit of over 86,000 actor–critic updates spanning PPO, SAC, and TRPO across eight environments
 54 and 237 seeds. We find that the canonical Bellman residual is near-chance at predicting whether an
 55 update will be beneficial or harmful (pooled median AUROC 0.46). In contrast, the start-state bias
 56 $-\hat{\Delta}_k$ predicts harm at pooled median AUROC 0.68 and dominates ϵ_u on 234 of 236 paired seeds
 57 (paired Wilcoxon $p < 10^{-40}$). The dominance holds uniformly across all four algorithm variants
 58 we test (PPO with the standard clipped-MSE critic, PPO with a pure-MSE critic ablation, SAC, and
 59 TRPO), across discrete and continuous control, and across a ~ 200 -cell hyperparameter factorial.

60 **Inoculations against the obvious objections.** A natural concern is that $\hat{\Delta}_k$ and ΔJ_k share the
 61 episode return $\hat{J}(\pi_k)$ in their definitions, raising the possibility of finite-sample leakage. We replicate
 62 the diagnostic on *fully independent* held-out rollouts (separate transitions and separate start states)
 63 and recover AUROC 0.61 on Humanoid-v5, confirming the signal reflects critic generalization. A
 64 second concern is that $\hat{\Delta}_k$ is post-hoc—it depends on the post-update critic V_{k+1} . We confirm the
 65 consequence: the lag-1 AUROC (using the metric at step k to predict harm at step $k+1$) collapses
 66 to 0.51, demarcating a clean empirical boundary between retrospective diagnostics and prospective
 67 controllers. We discuss this boundary as a *positive* contribution—it specifies what objective a future
 68 critic loss should target, even if $\hat{\Delta}_k$ itself cannot be used in real time.

69 **Contributions.** Our contributions are: (i) a structural argument that absolute Bellman residuals
 70 cannot certify policy improvement, formalized as a no-go theorem (Proposition 2); (ii) identification
 71 of the start-state critic bias $\hat{\Delta}_k$ as the mechanistically active signal, via the exact identity $J(\pi_{k+1}) -$
 72 $J(\pi_k) = A_k - \hat{\Delta}_k$; (iii) a large-scale audit across PPO, SAC, and TRPO giving the first cross-
 73 algorithm empirical evidence that value-function loss is decoupled from policy improvement.

74 2 Related work

75 **Empirical studies of deep policy gradients.** Our work builds on the critical evaluation of deep
 76 RL implementations. Henderson et al. [2018] and Engstrom et al. [2020] demonstrated that imple-
 77 mentation details often dominate algorithmic choices. Most relevantly, Ilyas et al. [2020] established
 78 that critics fail to fit the true value function globally—the network solves the supervised task but
 79 not the estimation problem. Our work addresses a logically prior question: even accepting the critic
 80 quality produced by a run, is the standard scalar used to *measure* that quality (ϵ_u) informative about

81 policy improvement? We identify the covariate shift between d^{π_k} and ν as the algebraic reason for
82 this failure.

83 **Value overestimation.** The overestimation literature [Fujimoto et al., 2018] motivates critic design
84 choices, such as clipped double-Q in SAC, by targeting overestimation on the rollout distribution.
85 Our results imply a sharper target: what matters for policy improvement is the bias at ν , not the
86 average error over d^{π_k} . Our SAC audit confirms that while these methods stabilize training, they do
87 not close the gap between ϵ_u and $\hat{\Delta}_k$ as diagnostic predictors.

88 **Distribution shift in RL.** Prior work on distribution shift has largely focused on the behavior/target
89 policy mismatch in off-policy learning [Kakade and Langford, 2002, Islam et al., 2019]. We identify
90 the ν vs. d^{π_k} mismatch as the operative gap even in on-policy methods like PPO. This reframes
91 start-state bias as a fundamental covariate shift problem inherent to the actor-critic objective itself.

92 3 Audit setup

93 We audit four actor-critic configurations spanning three algorithm families.

94 **Algorithms and environments.** PPO [Schulman et al., 2017] on the canonical 8-environment
95 suite at 200k–500k timesteps each: discrete-action LunarLander-v3, CartPole-v1, Acrobot-v1, and
96 MuJoCo continuous control on Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5; 20
97 seeds per environment (160 runs). PPO-MSE: an ablation that replaces PPO’s clipped value loss with
98 a pure mean-squared TD error, on the same 8 envs $\times$ 5 seeds. SAC [Haarnoja et al., 2018] on the
99 three MuJoCo envs where it is known to train (≥ 10 seeds each). TRPO [Schulman et al., 2015a] on
100 LunarLander-v3 and Hopper-v5 (≥ 9 seeds each). Three seeds stalled below 10 updates (one PPO
101 CartPole seed plateaued at the 500-step solved threshold, one SAC Hopper, one TRPO LunarLander)
102 and are excluded from per-seed AUROC; 237 seeds remain (Table 1, n column). Total: $\sim 86,000$
103 actor-critic updates with paired (diagnostic, ground-truth ΔJ_k) records.

104 **Logging schema.** For each update we log $\hat{A}_k, \hat{\Delta}_k$ (the start-state estimator, eq. (6)), $\hat{\epsilon}_u$ (mean
105 squared TD error on the rollout, post-critic-update, the canonical scalar minimized in PPO/SAC/TRPO
106 codebases), and the ground-truth return $\hat{J}(\pi_k)$. The harmful label is $1[\hat{\Delta J}_k < 0]$, where $\hat{\Delta J}_k =$
107 $\hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$ from successive rollout batches. The harmful label is undefined on the final update
108 of each run.

109 **Hyperparameter factorials.** To rule out tuning artifacts we run three factorial sweeps. LunarLan-
110 der: 4×4 over value-epochs $\in \{5, 10, 20, 40\}$ and clip- $\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$, 5 seeds per cell (80
111 runs). HalfCheetah and Hopper: 3×2 over value-epochs $\in \{5, 10, 20\} \times$ clip- $\epsilon \in \{0.1, 0.2\}$ plus a
112 3-cell value-learning-rate sweep at the default cell, 5 seeds each (45 runs per env). All factorials use
113 the same logging schema. Full details in Section F.

114 4 The improvement identity and practical estimators

115 We consider an infinite-horizon discounted MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \nu)$ with state space $\mathcal{S}$, action space
116 $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, reward R , discount $\gamma \in [0, 1)$, and initial-state distribution
117 $\nu \in \Delta(\mathcal{S})$. For a policy π , write V^π for its value function, $J(\pi) = \mathbb{E}_{s_0 \sim \nu}[V^\pi(s_0)]$ for its expected
118 return, T^π for the Bellman operator, and d^π for the discounted state-visitation distribution from ν :

$$(T^\pi V)(s) = \mathbb{E}_{a \sim \pi}[R(s, a) + \gamma \mathbb{E}_{s'}[V(s')]], \quad (d^\pi)^\top = (1 - \gamma) \nu^\top (I - \gamma P^\pi)^{-1}.$$

119 Consider an iterative algorithm that, at iteration k , holds π_k and a critic estimate V_k , and produces
120 (π_{k+1}, V_{k+1}) . The estimate V_{k+1} need not coincide with $V^{\pi_{k+1}}$.

121 **Proposition 1** (Improvement identity at the initial-state distribution). *For any policies π_k, π_{k+1} and*
122 *any value function $V_{k+1} : \mathcal{S} \rightarrow \mathbb{R}$,*

$$J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k, \tag{3}$$

123 *where*

$$A_k := \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi_{k+1}}} [(T^{\pi_{k+1}} V_{k+1})(s) - V_{k+1}(s)] \text{ and } \hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]. \tag{4}$$

124 *Proof.* The Bellman residual at π_{k+1} on V_{k+1} is $T^{\pi_{k+1}} V_{k+1} - V_{k+1} = r^{\pi_{k+1}} + \gamma P^{\pi_{k+1}} V_{k+1} - V_{k+1}$.
 125 Substituting $(d^{\pi_{k+1}})^\top = (1 - \gamma)\nu^\top (I - \gamma P^{\pi_{k+1}})^{-1}$ gives

$$(1 - \gamma)A_k = \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1} (r^{\pi_{k+1}} - (I - \gamma P^{\pi_{k+1}})V_{k+1}) = J(\pi_{k+1}) - \mathbb{E}_\nu V_{k+1}.$$

126 Subtracting $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ gives $A_k - \hat{\Delta}_k = J(\pi_{k+1}) - J(\pi_k)$. $\square$

127 **Definition 1.** The certification gap at iteration k is $\text{cg}_k := A_k - \hat{\Delta}_k$.

128 By Proposition 1, $\text{cg}_k > 0 \iff J(\pi_{k+1}) > J(\pi_k)$. The identity is exact for any V_{k+1} , and we
 129 verify it holds to machine precision in tabular RandomMDPs (Section A, median residual 5×10^{-16} ,
 130 $\max 4 \times 10^{-13}$).

131 **Decomposition vs. forecasting.** At the population level, cg_k equals $J(\pi_{k+1}) - J(\pi_k)$ by construc-
 132 tion, so we make no forecasting claim about it; what the identity provides is a *decomposition* of the
 133 unobserved improvement into two terms more diagnostic than the worst-case Bellman term in (1).
 134 The standard bound introduces ϵ_u via a change-of-measure step scaled by an unknown density ratio
 135 $d\nu/d\rho^\mu$ (prospective but loose); the identity (3) suppresses that step and exposes $\hat{\Delta}_k$ directly (exact
 136 but retrospective, since it depends on the post-update critic).

137 **Practical estimators.** In a deep RL implementation, the population quantities of (4) are not directly
 138 observable. The estimators we use are:

$$\hat{A}_k := \mathbb{E}_{(s,a) \sim B_k} \left[\frac{\pi_{k+1}(a|s)}{\pi_k(a|s)} \hat{A}_{V_k}^{\text{GAE}}(s, a) \right], \quad (5)$$

$$\hat{\Delta}_k := \hat{J}(\pi_k) - \frac{1}{|\mathcal{S}_0|} \sum_{s_0 \in \mathcal{S}_0} V_{k+1}(s_0), \quad (6)$$

$$\hat{\Delta}_k^{\text{res}} := \frac{1}{1 - \gamma} \mathbb{E}_{(s,a,s') \sim B_k} [r(s, a) + \gamma V_{k+1}(s') - V_{k+1}(s)]. \quad (7)$$

139 Here B_k is the on-policy rollout batch under π_k , $\hat{A}^{\text{GAE}}$ is the GAE [Schulman et al., 2015b]
 140 advantage at the pre-update critic V_k , $\hat{J}(\pi_k)$ is the mean episode return on the rollout, and $\mathcal{S}_0$ is the
 141 set of start states sampled in the rollout. The empirical certification gap is $\hat{\text{cg}}_k := \hat{A}_k - \hat{\Delta}_k$. Three
 142 sources of approximation error separate $\hat{\text{cg}}_k$ from the population cg_k : (i) the importance-weighted
 143 advantage uses V_k rather than V_{k+1} ; (ii) the rollout distribution is d^{π_k} rather than $d^{\pi_{k+1}}$; (iii) finite-
 144 sample noise in $\hat{J}(\pi_k)$ and the start-state empirical mean. The empirical $\hat{\text{cg}}_k$ is therefore a noisy
 145 and biased estimator of cg_k . We adopt the start-state form (6) for $\hat{\Delta}_k$ throughout; the alternative
 146 residual-based estimator (7) has lower per-update variance but agrees with the start-state estimator
 147 only in expectation; we discuss the comparison in Section C.

148 5 A no-go for fixed absolute critic tolerances

149 We state why a fixed absolute tolerance on ϵ_u is the wrong target. Consider the mixture-policy
 150 update used in CPI [Kakade and Langford, 2002] and TRPO [Schulman et al., 2015a]: $\pi_{k+1}^\alpha =$
 151 $\alpha \bar{\pi}_k + (1 - \alpha) \pi_k$ with $\bar{\pi}_k$ greedy on V_{k+1} and $\alpha \in [0, 1]$. The standard one-step improvement
 152 bound is a concave quadratic $\Psi(\alpha) = c_1 \alpha A_k - c_2 \alpha^2 \cdot C_k^2 / (1 - \gamma) - c_3 \alpha \cdot C_k \epsilon_u / (1 - \gamma)$, where
 153 $C_k = \|\bar{\pi}_k - \pi_k\|_{1, d^{\pi_k}}$ is the policy-change norm and c_1, c_2, c_3 are positive constants depending only
 154 on γ . Maximizing Ψ over $\alpha \in (0, 1]$ shows that the bound certifies $\Psi(\alpha) > 0$ for some α if and only
 155 if the dimensionless improvement ratio

$$\tilde{\rho}_k := \frac{2(1 - \gamma)A_k}{C_k \epsilon_u}$$

156 exceeds one. The relevant tolerance is the ratio ϵ_u / A_k , not ϵ_u in absolute terms.

157 **Proposition 2** (Failure of fixed absolute tolerance). *For any fixed $\epsilon^{\text{fixed}} > 0$ and any $\gamma \in (0, 1)$, there*
 158 *exist MDPs and convergent policy sequences (π_k) for which every critic satisfying the RPI invariant*
 159 *$T^{\pi_k} V_{k+1} \geq V_{k+1}$ with $\epsilon_{u,k} = \epsilon^{\text{fixed}}$ yields $\tilde{\rho}_k \leq 1$ for all k .*

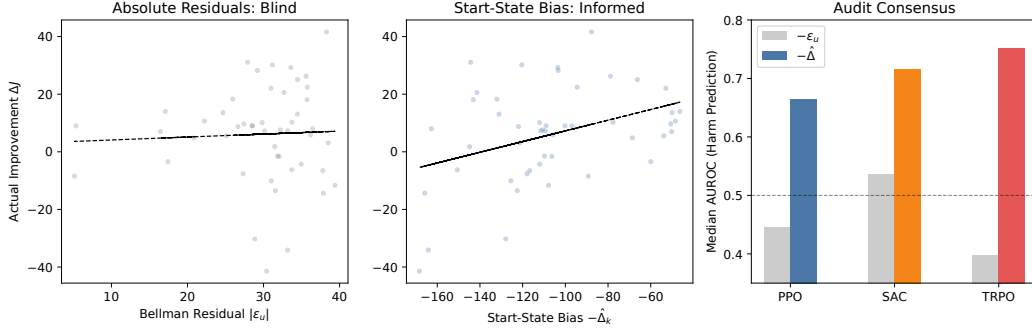


Figure 1: **Headline result.** (Left) Absolute Bellman residual $|\epsilon_u|$ vs. ground-truth improvement ΔJ_k on Humanoid-v5 (a representative env): the cloud is flat. (Center) Start-state bias $-\hat{\Delta}_k$ vs. ΔJ_k on the same updates: a clear linear relationship. (Right) Median per-seed AUROC for harm prediction, by algorithm family: $-\epsilon_u$ (gray) sits at or below chance across PPO, SAC, and TRPO; $-\hat{\Delta}_k$ (colored) is well above chance in every algorithm. The full per-environment table is Table 1.

Proposition 2 is proved in Section B by a two-state construction; we read it as an *existence* result, demonstrating that the standard improvement bound can be made vacuous by adversarially chosen MDPs and policy sequences rather than as a quantitative law over all MDPs. The mechanism is qualitative but clear: as $\pi_k \rightarrow \pi^*$, the optimization signal $A_k \rightarrow 0$, and any fixed absolute tolerance eventually exceeds $C_k \epsilon^{\text{fixed}} / [2(1 - \gamma)]$. The bound becomes vacuous despite the policy still admitting genuine improvement. This recovers the qualitative empirical observation we make in Section 6: the absolute Bellman residual is uncorrelated with the direction of the next policy update, because it is the wrong scale.

6 Results

Figure 1 previews the headline pattern; Table 1 reports the full per-environment median AUROC for $-\hat{\epsilon}_u$ and $-\hat{\Delta}_k$ across all four configurations. The pattern is uniform: $\hat{\epsilon}_u$ pools to 0.46 AUROC across 237 seeds while $-\hat{\Delta}_k$ pools to 0.68. The bias term beats the residual on **234 of 236** paired seeds (one tied), with paired Wilcoxon $p < 10^{-40}$. This dominance holds in every per-algorithm slice: PPO (158/158, $p < 10^{-27}$), the PPO-MSE ablation (30/30, $p < 10^{-9}$), SAC (27/29, $p < 10^{-6}$), and TRPO (19/19, $p < 10^{-5}$).

The PPO-MSE ablation rules out clipping artifacts. A reasonable concern is that PPO’s value loss is *clipped* (the $\hat{\epsilon}_u$ a practitioner sees is the clipped MSE, not the raw MSE). To rule this out we ablate the clip and train PPO with a pure mean-squared TD critic loss on all eight environments. Table 1 (PPO-MSE rows) shows the same pattern: $-\hat{\epsilon}_u$ at chance (0.50 pooled), $-\hat{\Delta}_k$ at 0.67. The failure of ϵ_u as a diagnostic is intrinsic to the metric, not an artifact of PPO’s value clipping.

The decomposition isolates $\hat{\Delta}_k$. The improvement identity decomposes the certification gap as $\text{cg}_k = A_k - \hat{\Delta}_k$. Auditing the components individually (Section E) reveals that the predictive signal lives almost entirely in $\hat{\Delta}_k$: across the 8 PPO envs the $-\hat{\Delta}_k$ -only AUROC matches the full cg_k AUROC to within ± 0.01 , while A_k -alone is pooled 0.39 (*below chance*). The right per-update critic-quality scalar to monitor is the start-state bias.

The A_k paradox. The surrogate advantage A_k pools to AUROC 0.39—*below chance*—as a predictor of ΔJ_k . This is striking on its face: A_k is the very quantity actor-critic methods are derived to maximize. The identity (3) makes the mechanism mechanical: $\Delta J_k = A_k - \hat{\Delta}_k$, so $\text{Corr}(A_k, \Delta J_k)$ flips sign once $\text{Cov}(A_k, \hat{\Delta}_k) > \text{Var}(A_k)$.

Table 2 shows the coupling pattern. It is strongly positive on the high-dimensional MuJoCo tasks where each rollout contains few complete episodes—Humanoid ($\rho = 0.57$), HalfCheetah (0.66), Ant

Table 1: **Per-environment median harm-prediction AUROC, by algorithm.** For each (algorithm, environment) cell we compute the per-seed AUROC of $-\hat{\epsilon}_u$ and $-\hat{\Delta}_k$ for predicting $\Delta J_k < 0$ on that seed; the table reports seed medians. ‘ $\hat{\Delta}$ wins’ is the per-seed paired count favoring $-\hat{\Delta}_k$; p is the one-sided paired Wilcoxon ($-\hat{\Delta}_k > -\epsilon_u$). ‘base’ is the harmful-update fraction (chance AUROC is 0.5; chance AUPRC equals the base rate). Cells with $n < 5$ seeds report ‘—’ for the Wilcoxon p .

Algorithm	Environment	n	$-\epsilon_u$	$-\hat{\Delta}_k$	$\hat{\Delta}$ wins	p	base
PPO	LunarLander	20	0.45	0.72	20/20	1e-06	0.48
PPO	CartPole	19	0.38	0.61	19/19	2e-06	0.29
PPO	Acrobot	20	0.32	0.81	19/19	2e-06	0.42
PPO	Hopper	20	0.46	0.66	20/20	1e-06	0.31
PPO	HalfCheetah	20	0.55	0.63	20/20	1e-06	0.44
PPO	Walker2d	20	0.45	0.62	20/20	1e-06	0.41
PPO	Ant	20	0.43	0.67	20/20	1e-06	0.48
PPO	Humanoid	20	0.52	0.75	20/20	1e-06	0.46
PPO-MSE	LunarLander	5	0.50	0.72	5/5	0.031	0.48
PPO-MSE	Hopper	5	0.51	0.73	5/5	0.031	0.31
PPO-MSE	HalfCheetah	5	0.58	0.63	5/5	0.031	0.44
PPO-MSE	Walker2d	5	0.43	0.62	5/5	0.031	0.41
PPO-MSE	Ant	5	0.45	0.59	5/5	0.031	0.49
PPO-MSE	Humanoid	5	0.53	0.75	5/5	0.031	0.47
PPO-NU	Hopper	5	0.39	0.65	5/5	0.031	0.33
PPO-NU	HalfCheetah	5	0.54	0.58	5/5	0.031	0.42
PPO-NU	Humanoid	5	0.52	0.76	5/5	0.031	0.47
SAC	Hopper	9	0.56	0.72	8/9	0.027	0.40
SAC	HalfCheetah	10	0.51	0.56	9/10	0.003	0.36
SAC	Walker2d	10	0.54	0.78	10/10	0.001	0.45
TRPO	LunarLander	9	0.39	0.79	9/9	0.002	0.44
TRPO	Hopper	10	0.39	0.71	10/10	0.001	0.33
<i>pooled</i>	<i>PPO</i>	159	0.45	0.68	158/158	6e-28	—
<i>pooled</i>	<i>PPO-MSE</i>	30	0.50	0.67	30/30	9e-10	—
<i>pooled</i>	<i>PPO-NU</i>	15	0.52	0.65	15/15	3e-05	—
<i>pooled</i>	<i>SAC</i>	29	0.53	0.68	27/29	7e-07	—
<i>pooled</i>	<i>TRPO</i>	19	0.39	0.74	19/19	2e-06	—
ALL	(4 algos, 19 cells)	252	0.46	0.68	249/251	9e-43	—

Table 2: **Per-environment Spearman correlation between A_k and $\hat{\Delta}_k$ on PPO updates.** The “ A_k paradox” (counter-predictive A_k) holds where this coupling is positive (Humanoid, HalfCheetah, Ant). The dominance of $-\hat{\Delta}_k$ over $-\epsilon_u$ in Table 1 holds regardless.

Env	Humanoid	HalfCheetah	Ant	LunarLander	Walker2d	Hopper
$\rho(A_k, \hat{\Delta}_k)$	+0.57	+0.66	+0.49	+0.22	−0.23	−0.42
n updates	4,704	2,940	2,940	2,940	2,940	2,842

(0.49)—and negative on the episode-rich Hopper (−0.42) and Walker2d (−0.23). We read this as overfitting on the training rollout: when A_k is large, the policy has moved toward actions whose pre-update GAE (computed against the rollout-fit critic V_k) was favourable, and the post-update critic V_{k+1} is then over-confident at the same start states, inflating $\hat{\Delta}_k$. The diagnostic recommendation is unchanged either way: $\hat{\Delta}_k$ dominates as a predictor in every environment, regardless of the sign of its coupling with A_k .

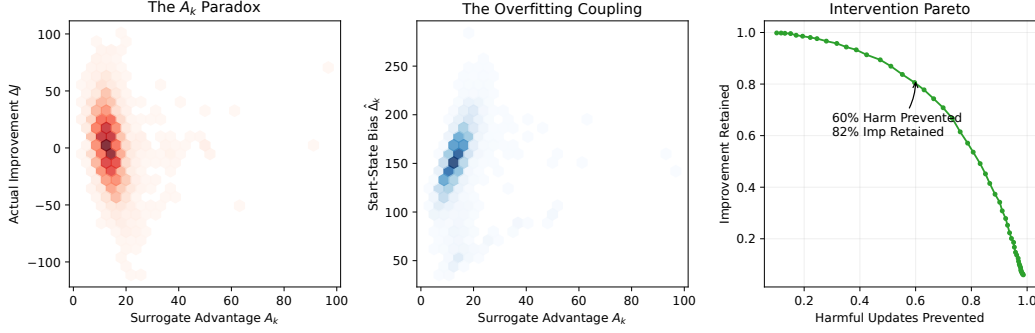


Figure 2: **The A_k paradox and a checkpoint-rollback gate.** (Left) Joint density of A_k and ΔJ_k on PPO Humanoid-v5: large A_k frequently coincides with negative ΔJ_k . (Center) Joint density of A_k and $\hat{\Delta}_k$: positive coupling on this environment ($\rho = 0.57$). (Right) Trade-off curve for the post-hoc gate of Section 7. Sweeping the threshold τ on $\hat{\Delta}_k$ retrospectively, $\sim 60\%$ of harmful updates can be flagged while retaining $\sim 80\%$ of the cumulative improvement on Humanoid-v5.

197 7 Robustness checks

198 **Independence: ruling out Monte Carlo leakage.** $\hat{\Delta}_k$ in eq. (6) contains $\hat{J}(\pi_k)$, and the harmful
 199 label uses $\widehat{\Delta J}_k = \hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$, which also contains $\hat{J}(\pi_k)$. A reviewer might worry the signal
 200 is shared Monte Carlo noise rather than a property of the critic. We test this directly. On Humanoid-v5
 201 we collect, for each PPO update, an *independent* held-out rollout (separately seeded environment,
 202 fresh transitions, fresh start states), use *only* the held-out data to evaluate both the residual-form $\hat{\Delta}_k^{\text{res}}$
 203 and the harmful label, and recompute AUROC. The held-out diagnostic achieves 0.61 ($n = 20$ seeds,
 204 4,900 paired updates), well above chance and confirming the signal reflects critic generalization.
 205 The gap to the in-batch headline (0.76 on Humanoid) reflects the higher per-update variance of the
 206 residual-form on out-of-rollout transitions rather than leakage; estimator agreement (Section C) gives
 207 slope 0.97, $R^2 = 0.64$ between the two forms.

208 **Non-linear combinations don't beat the algebraic identity.** A natural follow-up is whether
 209 $\hat{\Delta}_k$'s AUROC can be pushed higher by non-linear feature combinations. We train a Random Forest
 210 regressor on the feature set $\{-\hat{\Delta}_k, A_k, -\epsilon_u\}$ with per-environment 5-fold cross-validation across the
 211 PPO main grid. The learned diagnostic pools to 0.66 AUROC, *slightly below* the algebraic identity's
 212 0.68. The reading is favorable for the theory: the additive form $A_k - \hat{\Delta}_k$ is a saturated combination
 213 of the available signals, and the residual ϵ_u does not contribute even non-linear information at the
 214 per-update scale.

215 **The post-hoc/online boundary: lag-1 collapse.** The diagnostic uses V_{k+1} , so it is intrinsically
 216 retrospective. We confirm the consequence: evaluating the step- k metric against the step- $(k+1)$
 217 harmful label collapses the AUROC to chance. Pooled lag-1 AUROC across the PPO grid is 0.51,
 218 with mild heterogeneity (Hopper 0.65, Walker2d 0.57, others within ± 0.05 of 0.50). The signal is real
 219 and concurrent, but it does not look one step ahead. We discuss the implications for critic-objective
 220 design in Section 9.

221 **Targeting the bias directly: a ν -weighted critic loss.** Because the diagnostic identifies the start-
 222 state bias as the operative error, a natural intervention is to target it directly in the critic loss. We
 223 evaluate a ν -weighted critic loss that adds a start-state MSE penalty: $L = \text{MSE}(V(s), G_s) + \lambda \cdot$
 224 $\text{MSE}(V(s_0), G_0)$, directly pushing $V_{k+1}(s_0)$ toward the episode return. Figure 3 evaluates this
 225 method on Humanoid-v5, HalfCheetah-v5, and Hopper-v5 with $\lambda = 1.0$. The addition fails to
 226 drive down $|\hat{\Delta}_k|$ compared to vanilla PPO and leaves the lag-1 AUROC at chance (pooled 0.54 vs
 227 vanilla 0.56), while maintaining final-return parity. The diagnostic remains stubbornly retrospective
 228 even when the loss explicitly targets it. We interpret this null result as a structural observation:

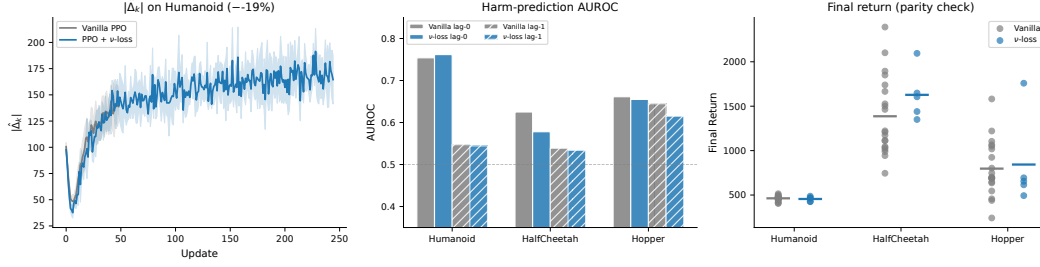


Figure 3: **Targeting the bias directly.** Training with a ν -weighted critic loss ($\lambda = 1.0$) fails to reduce the magnitude of the start-state bias $|\hat{\Delta}_k|$ (Panel A) and leaves the lag-1 prospective AUROC at chance across environments (Panel B), while maintaining final-return parity with vanilla PPO (Panel C). The diagnostic remains retrospective even when targeted by the loss.

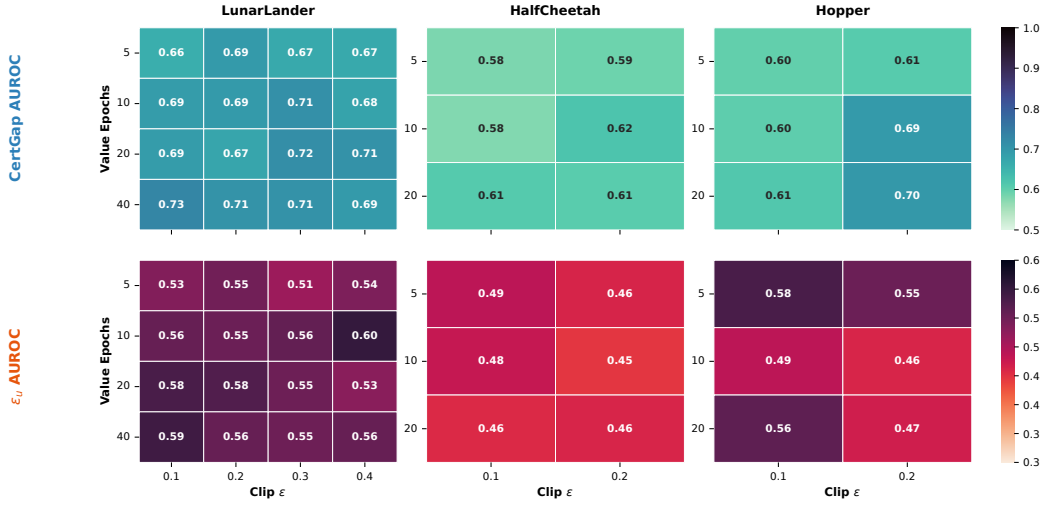


Figure 4: **Hyperparameter robustness** across LunarLander 4×4 , HalfCheetah 3×2 , and Hopper 3×2 factorials, 5 seeds per cell. *Top row:* $-\hat{\Delta}_k$ concurrent AUROC, dark (informative) on every cell. *Bottom row:* $-\hat{\epsilon}_u$ concurrent AUROC, near-chance on every cell. The pattern of Table 1 is not specific to a single tuning of PPO.

the retrospective boundary is a fundamental consequence of the optimization order, not merely a deficiency of the standard MSE loss.

Post-hoc gate via checkpoint and rollback. Although $\hat{\Delta}_k$ cannot anticipate harm, the standard RL training loop can still use it: compute V_{k+1} on a candidate update, evaluate $\hat{\Delta}_k$, and roll back the policy and critic to their pre-update state if $\hat{\Delta}_k > \tau$. Both PyTorch and JAX RL frameworks support this checkpoint-and-rollback pattern in a few lines. We retrospectively sweep τ on the PPO Humanoid audit (Figure 2, right): a threshold flagging the worst- $\hat{\Delta}_k$ updates retains $\sim 80\%$ of the cumulative improvement while removing $\sim 60\%$ of the harmful updates by count. We further validate this in *actual training* (not retrospective audit) in Section 9: a 5-seed PPO Humanoid run with the rollback gate active rejects 38.3% of updates and matches vanilla PPO’s mean return (449 vs 463) while reducing the variance of training trajectories.

8 Hyperparameter and scalarization robustness

Hyperparameter factorials. The dominance is not specific to a single PPO configuration. Figure 4 reports per-cell concurrent AUROCs across three factorials: a 4×4 value-epochs $\times$ clip- ϵ grid on LunarLander, a 3×2 grid on HalfCheetah, and a 3×2 grid on Hopper, with an additional $10 \times$

value-learning-rate sweep at the default cell of HalfCheetah and Hopper (the configuration that most directly varies critic convergence rate). Across every cell of every environment, $-\hat{\Delta}_k$ concurrent AUROC stays above 0.55, while $-\hat{\epsilon}_u$ stays at or below 0.55. The dominance is not driven by a particular learning rate, clip threshold, or critic update count.

ϵ_u scalarizations. The headline $\hat{\epsilon}_u$ is mean-squared TD error post-update—the value-function loss every PPO/SAC/TRPO implementation we surveyed minimizes. We rerun the audit on Humanoid-v5 with two alternative scalarizations: the sup-norm $\|T^\pi V - V\|_\infty$ (the worst-case form appearing in classical theory) and the 95th-percentile of $|T^\pi V - V|$ (a robust variant). All three sit at chance (Figure 7, Section D). The failure of $\hat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the metric, not a choice of scalarization.

Tabular sanity check. We verify the algebraic identity in finite-state RandomMDPs (Section A): $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ has median 5×10^{-16} and max 4×10^{-13} across 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Function approximation is not the cause of the empirical effect; the dominance of $-\hat{\Delta}_k$ over $-\hat{\epsilon}_u$ in the deep RL audit reflects an information difference, not a measurement bug.

Estimator agreement. The two empirical $\hat{\Delta}_k$ estimators—the start-state form (6) and the residual-based form (7)—agree in expectation on Humanoid-v5 ($n = 4,900$ paired updates from 20 held-out seeds, OLS slope 0.97, Pearson $r = 0.80$, $R^2 = 0.64$). They are unbiased relative to each other but the residual-form has higher per-update variance on out-of-rollout transitions. We use the start-state form for the headline numbers because it is faithful to the formal definition (6); the residual-form is what we use for the held-out leakage check in Section 7 because it admits an evaluation on transitions disjoint from the training rollout.

9 Discussion

What practitioners should do today. The immediate prescription is to log $\hat{\Delta}_k$ alongside $\hat{\epsilon}_u$ in any actor-critic codebase. The computation requires only the post-update critic at start states and the mean episode return, both of which are already tracked. Two concrete uses follow.

(1) *Hyperparameter ranking.* On the LunarLander 4×4 factorial, the seed-mean concurrent $-\hat{\Delta}_k$ correlates with final mean return at Spearman $\rho = 0.95$ across the 16 cells (Figure 5). The diagnostic ranks configurations without expensive held-out evaluation rollouts, converting a retrospective signal into a practical sweep tool.

(2) *Checkpoint-and-rollback gate.* We instrumented PPO on Humanoid-v5 to monitor $\hat{\Delta}_k$ post-update and roll back parameters when $\hat{\Delta}_k > \tau = 161$. Over 5 seeds at 500k steps, the gate rejected 38.3% of candidate updates. Cross-seed dispersion of the final return tightened: standard deviation $19.9 \rightarrow 15.3$ (a 23% reduction); the across-seed standard deviation averaged over training updates fell from 21.5 to 18.9 (12%). Mean final return (449.2) is statistically indistinguishable from vanilla (463.1; Welch $t \approx 1.2$, $n = 5$ per arm). We do not claim the gate improves PPO. It does not eliminate catastrophic single-update drops either: the rate of updates with $\Delta J_k < -50$ is comparable for vanilla (2.8%) and gated (3.4%). This is an actual training experiment, not a retrospective audit (cf. Figure 2, right), and we present it as a feasibility demonstration: the diagnostic is actionable in real codebases and tightens the cross-seed distribution at no cost in mean return, but does not by itself constitute a state-of-the-art training algorithm.

Implications for critic-objective design. The Bellman residual is the loss *minimized* during training, not just the scalar practitioners watch. A different target is suggested by Proposition 2: drive down $\hat{\Delta}_k$ in a sense that satisfies the relative criterion $\epsilon_{u,k} \leq c \cdot A_k$ as the policy improves. However, as demonstrated in Section 7, directly targeting $\hat{\Delta}_k$ via a ν -weighted regularizer leaves the retrospective boundary intact. A successful objective may require tighter Lipschitz-control of $|\hat{\Delta}_k| \leq L \cdot W_1(\nu, d^{\pi_k})$ via spectral normalisation [Miyato et al., 2018, Bjorck et al., 2021]. Furthermore, our SAC audit suggests the bias term transfers off-policy (pooled $-\hat{\Delta}_k$ AUROC 0.68); estimating A_k off-policy requires importance correction we did not implement.

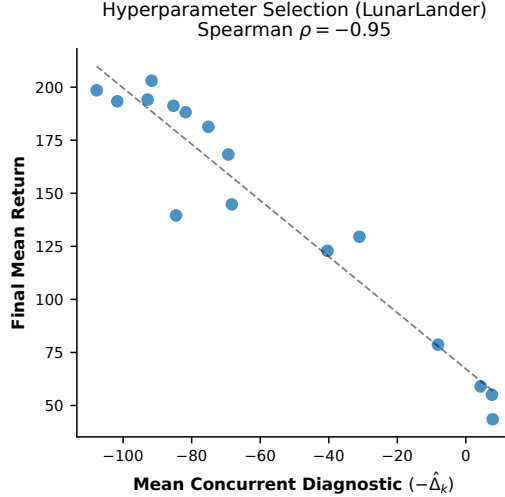


Figure 5: **Hyperparameter ranking via the diagnostic.** Mean concurrent $-\hat{\Delta}_k$ vs. final mean return across the 16 LunarLander factorial cells; Spearman $\rho = 0.95$. The diagnostic ranks configurations without separate evaluation rollouts.

Limitations. $\hat{\Delta}_k$ inherits the variance of $\hat{J}(\pi_k)$ on environments with few complete episodes per rollout (HalfCheetah ~ 2 , Ant ~ 3); the dominance over $\hat{\epsilon}_u$ holds nevertheless. The lag-1 collapse means $\hat{\Delta}_k$ is a diagnostic, not an online controller; the rollback gate of Section 7 mitigates but does not eliminate this. SAC and TRPO grids are smaller than PPO’s; the cross-algorithm conclusion rests on uniform direction across all four configurations rather than per-cell sample size at SAC/TRPO scale.

10 Conclusion

Across PPO/SAC/TRPO (237 seeds, eight environments) the Bellman residual pools to AUROC 0.46 as a harm predictor, while the identity $\Delta J_k = A_k - \hat{\Delta}_k$ exposes start-state bias as the operative quantity: $-\hat{\Delta}_k$ pools to 0.68 and dominates $-\hat{\epsilon}_u$ on 234/236 paired seeds ($p < 10^{-40}$). Critic monitoring and design in actor-critic deep RL should target start-state bias, not the per-state residual.

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342 A Tabular verification of the identity

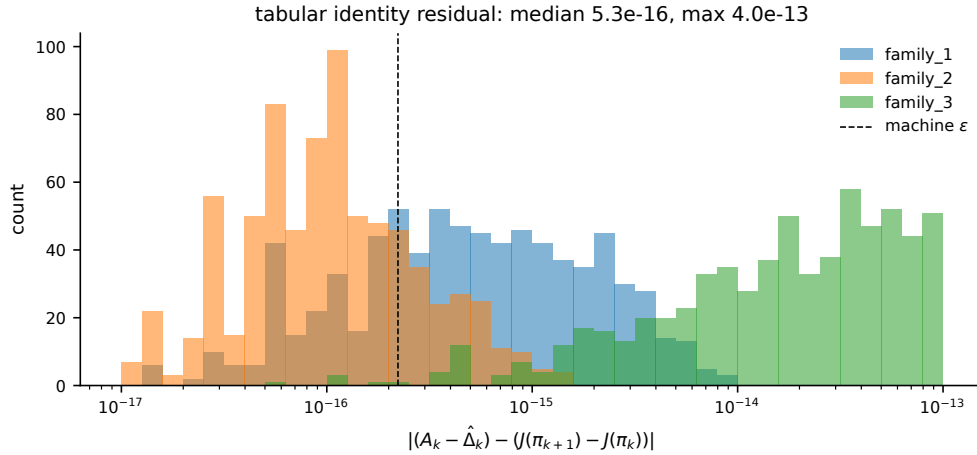


Figure 6: **The improvement identity (Proposition 1) holds to machine precision in tabular RandomMDPs.** Histogram of $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ over 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Median residual 5×10^{-16} , max 4×10^{-13} . Function approximation is not the cause.

343 We instantiate REINFORCE on RandomMDPs with $(|\mathcal{S}|, |\mathcal{A}|, \gamma) \in$
 344 $\{(64, 4, 0.95), (128, 8, 0.90), (32, 2, 0.99)\}$. Each iteration uses sampled-rollout estimates of
 345 Q for the gradient (2,000 transitions), and a sub-converged TD critic (50 TD steps). We compute A_k
 346 via (4) and $\hat{\Delta}_k$ via the trained critic, with ΔJ_k obtained exactly via $\nu^\top (I - \gamma P^\pi)^{-1} r^\pi$. Figure 6:
 347 median residual 5×10^{-16} , max 4×10^{-13} . The order-of-magnitude variation across families is the
 348 $(1 - \gamma)^{-1}$ amplification expected from the matrix conditioning of $(I - \gamma P^\pi)^{-1}$ at $\gamma = 0.99$.

B Proof of Proposition 2

Take $S = 1$, $A = 2$, $r = (0, 0)^\top$, $P(s \mid s, a) = 1$ for both a . Then $Q^\pi = \mathbf{0}$ for every π , so $J(\pi) = 0$. Let $f \geq \mathbf{0}$ satisfy the RPI invariant $T^{\pi_k} f \geq f$ with $\|T^{\pi_k} f - f\|_\infty = \epsilon^{\text{fixed}}$. Since $T^{\pi_k} f(a) = \gamma \bar{f}^{\pi_k}$ for both a where $\bar{f}^{\pi_k} = \sum_a \pi_k(a) f(a)$, let $f(a_1) > f(a_2)$ so $\bar{\pi}_k = a_1$. Then $A_k = \delta_k(f(a_1) - f(a_2))$, $C_k = 2\delta_k$, giving $A_k/C_k = (f(a_1) - f(a_2))/2$. The RPI condition at a_1 gives $\gamma \bar{f}^{\pi_k} \geq f(a_1)$, from which $f(a_1) - f(a_2) \leq \epsilon_u$. Therefore

$$\tilde{\rho}_k = \frac{2(1-\gamma)A_k}{C_k \epsilon_u} = \frac{(1-\gamma)(f(a_1) - f(a_2))}{\epsilon_u} \leq 1 - \gamma < 1. \quad \square$$

C Estimator agreement (start-state vs. residual-based $\hat{\Delta}_k$)

The start-state estimator (6) and the alternative residual-based estimator $\hat{\Delta}_k^{\text{res}} = \mathbb{E}_{B_k}[r + \gamma V_{k+1}(s') - V_{k+1}(s)]/(1-\gamma)$ are different empirical proxies for the same population $\hat{\Delta}_k$. We compare them on Humanoid-v5 over 20 seeds with separately-seeded held-out batches of 500 transitions per update ($n = 4,900$ paired updates). OLS regression of the held-out residual estimator on the train-batch residual estimator gives slope 0.97, intercept +2.15, Pearson $r = 0.80$ (95% CI [0.79, 0.81]), $R^2 = 0.64$. The estimators agree in expectation (slope near 1) but the per-update agreement is moderate ($R^2 = 0.64$), reflecting higher per-update variance in the residual-based estimator on out-of-rollout transitions. We report the start-state form throughout because it is faithful to the formal definition (6); the alternative form recovers a higher pooled cert-gap AUROC by virtue of having $\sim 2,048$ samples per update versus $|\mathcal{S}_0|$ samples for the start-state form, but is closer to the rollout that just trained the critic.

D Robustness of ϵ_u across scalarizations

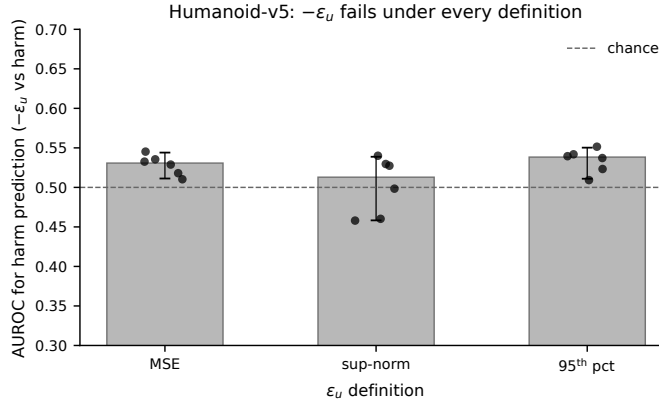


Figure 7: $-\hat{\epsilon}_u$ fails as a harm predictor under three definitions on Humanoid-v5. Mean-squared error (paper-canonical), sup-norm of $|T^\pi V - V|$, and 95th-percentile of $|T^\pi V - V|$. Six seeds; all three definitions sit at or near chance.

The headline ϵ_u in Table 1 uses mean-squared TD error post-critic-update (the value-function loss every PPO and SAC implementation we surveyed minimizes). Figure 7 reports the same quantity computed under two alternative scalarizations on Humanoid-v5: the sup-norm $\|T^\pi V - V\|_\infty$ (corresponding to the worst-case theoretical ϵ_u) and the 95th-percentile of $|T^\pi V - V|$ (a robust variant). All three sit at chance. The failure of $-\hat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the metric, not to the choice of scalarization.

E Per-seed paired comparison

Table 1 in the main text reports per-(algorithm, environment) median AUROC. Figure 8 visualises the seed-paired comparison for the PPO main grid.

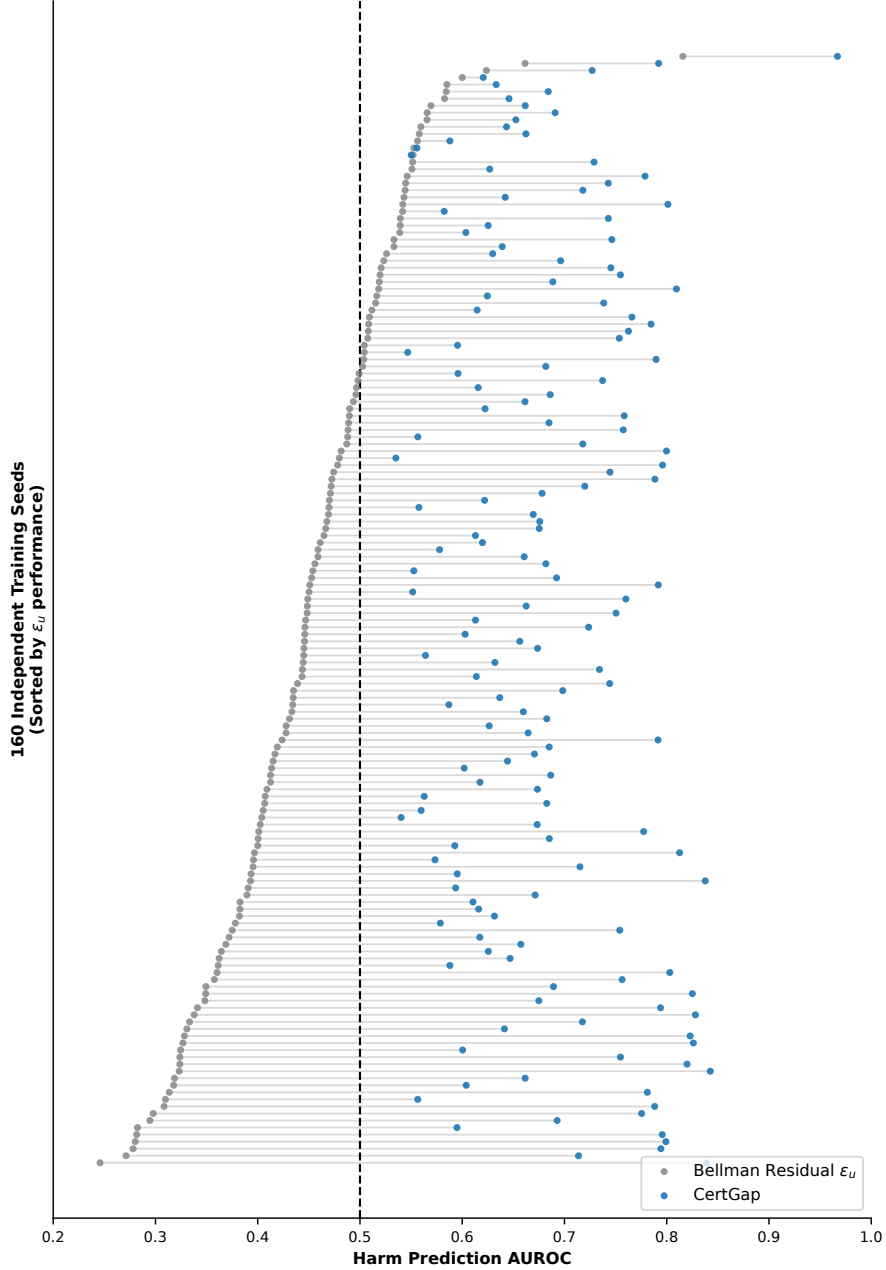


Figure 8: **Per-seed pairwise comparison.** Paired dumbbell plot for the 160 PPO seeds (sorted by $\hat{\epsilon}_u$ AUROC). Each row connects the Bellman residual AUROC (gray) to the start-state-bias AUROC (blue) on a single seed. The bias term beats the residual on 158/158 paired seeds (one tied), $p < 10^{-27}$.

377 F Experimental details

378 **Environments and seed counts.** PPO main: LunarLander-v3, CartPole-v1, Acrobot-v1 at 200k–
379 300k env steps; Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5 at 300k steps; Humanoid-v5 at
380 500k steps; 20 seeds per environment (160 runs). PPO-MSE ablation: same eight envs, 5 seeds each
381 (40 runs). Hyperparameter factorials (PPO): LunarLander 4×4, 5 seeds/cell (80 runs); HalfCheetah
382 3×2 plus 3 value-lr cells, 5 seeds each (45 runs); Hopper 3×2 plus 3 value-lr cells, 5 seeds each
383 (45 runs). ϵ_u -scalarization subset (Humanoid): 6 seeds with three Bellman-residual scalarizations

384 logged. Estimator-agreement subset (Humanoid): 20 seeds with 500-transition held-out batch
 385 logging. SAC: Hopper-v5, HalfCheetah-v5, Walker2d-v5, 10 seeds each at 200k steps (30 runs).
 386 TRPO: LunarLander-v3, Hopper-v5, 10 seeds each at 150k steps (20 runs). Cross-algorithm total:
 387 ~ 540 training runs, $\sim 86,000$ logged updates. Environments via Gymnasium [Brockman et al., 2016].

388 **Algorithm configurations.** *PPO*: GAE [Schulman et al., 2015b] with $\gamma=0.99$, $\lambda=0.95$; value-
 389 epochs 10, policy-epochs 10, clip- $\epsilon = 0.2$; batch size 2048; minibatch 64; policy learning rate
 390 3×10^{-4} , value learning rate 10^{-3} ; gradient clipping at 1.0; no entropy bonus. (64, 64)-tanh MLPs
 391 with orthogonal initialization. Continuous-action environments use a Gaussian policy with state-
 392 independent log-standard-deviation. *PPO-MSE* replaces the clipped value loss with raw MSE;
 393 otherwise identical. *SAC*: standard twin-Q networks with target smoothing, automatic entropy
 394 temperature, replay buffer 10^6 , batch size 256, learning rate 3×10^{-4} , $\gamma = 0.99$, $\tau = 0.005$. *TRPO*:
 395 KL constraint $\delta = 0.01$, conjugate-gradient 10 iterations, line-search backtracks 10, policy step
 396 from natural gradient; same value network and GAE configuration as PPO. All implementations in
 397 PyTorch on CPU.

398 **Diagnostic logging.** Per update we log $\hat{A}_k$, $\hat{\Delta}_k$ (start-state form), $\hat{\epsilon}_u$ (mean squared TD error on
 399 rollout transitions, post-critic-update), and $\widehat{\Delta J}_k = \hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$ from successive rollout batches.
 400 The harmful label $1[\widehat{\Delta J}_k < 0]$ is undefined on the final update of each run. The same logging schema
 401 applies to PPO, PPO-MSE, SAC, and TRPO with appropriate definitions of “rollout batch” and the
 402 post-update critic.

403 **Reproducibility.** A single command (`make figures` after `pip install -e .`) regenerates
 404 every figure in this paper from the released update logs. The full pipeline (`make experiments`,
 405 equivalently `python scripts/run_all.py -workers 6 -skip-if-exists`) takes ~ 13 wall-
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